CartNexus

Internship Progress Presentation

Men’s Fashion E-Commerce Platform

NextTech Limited · Department of CSE · Batch 26/220 · Spring 2026

Agenda Today

- Project & team
- Goals & progress
- Architecture & diagrams
- Storefront & admin features
- Database design
- Challenges & next steps
- Demo & Q&A

Team Contributors

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Project Overview Summary

CartNexus is a full-stack bilingual (English / Bangla) web store for men's fashion: browsing, cart, checkout, customer accounts, and an admin panel for catalog, orders, inventory, and content.

Goals

- Modern storefront UX
- REST API + MySQL persistence
- Admin dashboard & inventory

Status

Core MVP implemented — testing & deployment in progress.

Indicative completion (~78%)

Technology Stack Layers

Hover each layer — interactive overview.

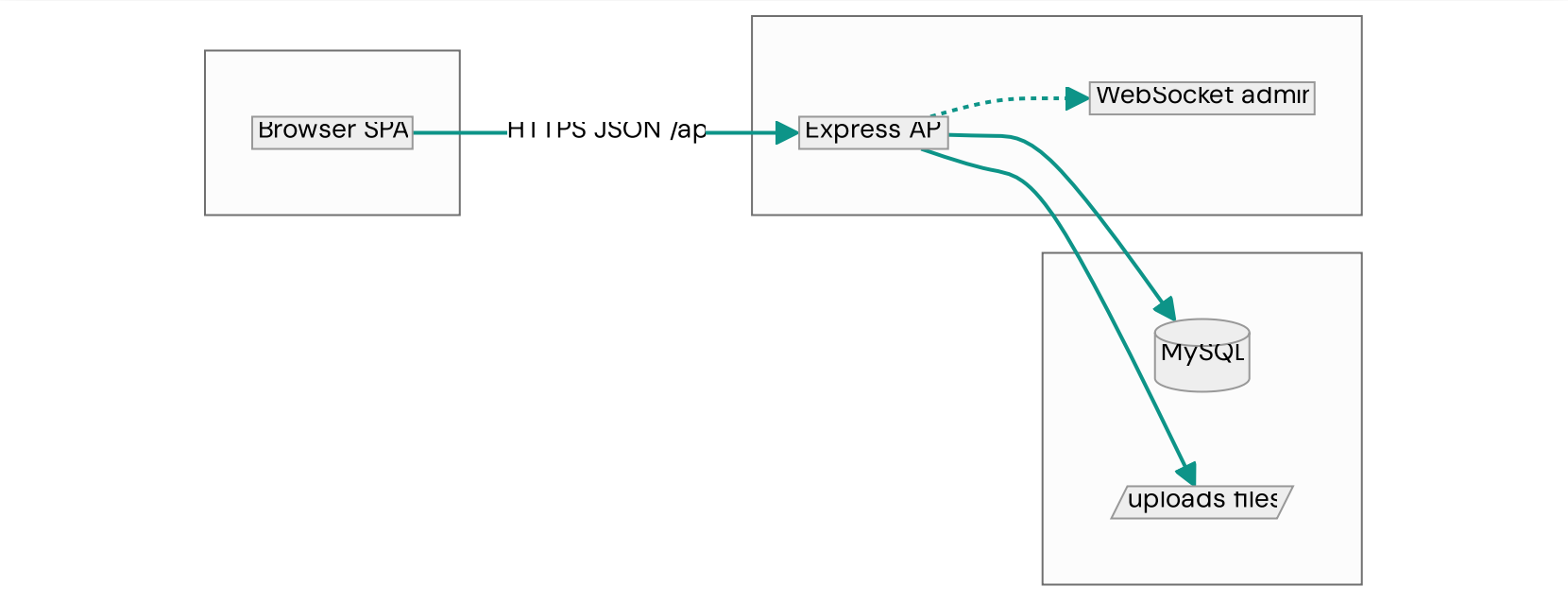
Presentation
React 18 · Vite · Tailwind CSS · React Router · Framer Motion

i18n & UX
i18next (EN/BN) · responsive layouts · Recharts (admin)

Application
Node.js · Express · JWT · bcrypt · Multer · WebSocket

Data
MySQL 8 · relational schema · migrations · static /uploads

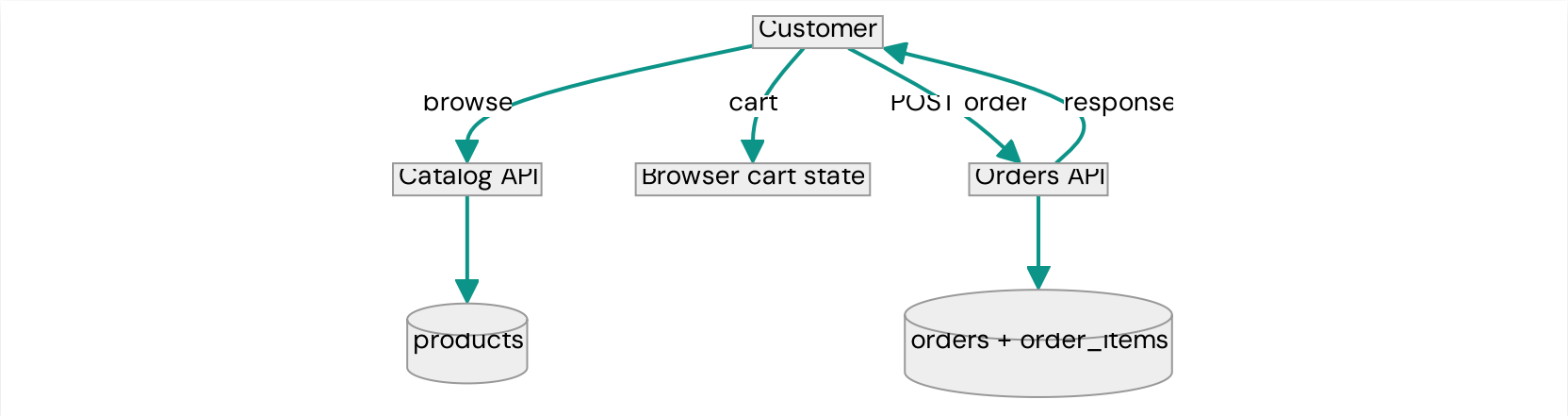
System Architecture 3-Tier



Production: reverse proxy + TLS · Dev: Vite proxy to API.

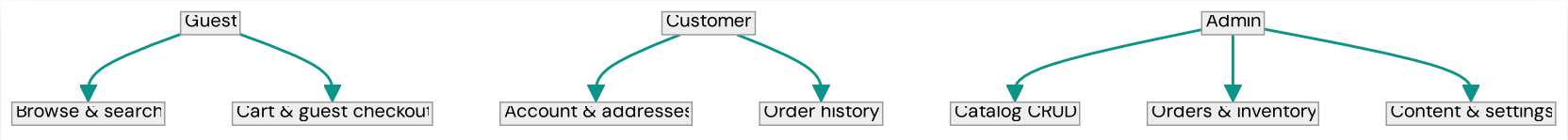
Data Flow

Checkout path



Actors & Features

Use cases



Storefront Delivered Customer

- Home hero banner (API-driven)
- Shop · categories / brands · product detail
- Color variants · cart context
- Checkout · guest or logged-in
- Blog · CMS pages (terms, privacy, FAQs)
- Footer · store settings · WhatsApp / Messenger FAB

Admin Panel

Back office

- Dashboard analytics & charts
- Products · categories · brands
- Orders · inventory & stock movements
- Users · home hero · blog · CMS
- Store settings (contact, social, chat links)

Database Design

ER core

Full DDL: `backend/db/schema.sql`

Challenges Mitigation

- **Schema drift:** migrations + changelog discipline
- **Deployment:** env secrets · HTTPS · persistent uploads
- **i18n:** UI + product fields in two languages
- **Payments:** COD baseline · gateway integration = future work

Next Steps Roadmap

1. End-to-end QA & bug fixes
2. Performance (Lighthouse) & smoke API tests
3. Production deploy & monitoring
4. SEO fine-tuning & analytics
5. Final documentation & demo recording

Thank you

Demo · Questions & Answers

F fullscreen · Esc slide overview · click or → to advance fragments

Speaker notes